

GOKTUG MERT OZDOGAN

Software Engineer | AI/ML and Backend Systems | Building AI-Driven Products

Istanbul, Turkey • goekmeroz@gmail.com • gokmeroz.com • github.com/gokmeroz • linkedin.com/in/goktugmertozdogan

PROFESSIONAL SUMMARY

Software Engineer specializing in **AI integrated full stack applications, backend systems, and cloud deployment**. Built **Nummoria**, a cross platform **AI powered personal finance SaaS** using **Node.js, React, React Native, MongoDB, and LLM APIs**, and **contributed to an agentic AI system for scientific paper reproducibility** through inzva AI Projects #10. Previously delivered production features supporting more than **10,000 employees at a major Turkish bank using C#/.NET Core and Angular**. Strong foundation in **Python, machine learning, REST APIs, system design, data structures, Docker, AWS**, and scalable backend architecture.

EXPERIENCE

Halkbank — One of Turkey's Largest State-Owned Banks

04/2025 – 07/2025

Full-Stack Software Engineering Intern

Istanbul, Turkey

- Implemented **Personnel Absence System** serving **10,000+ employees**, attendance-tracking logic in **C#/.NET Core** across **75+ relational tables** (holidays, medical leave, approval workflows).
- **Developed full stack reporting workflows** using **Angular and ASP.NET Core**, supporting real time data processing across **800+ bank branches** throughout Turkey.
- **Delivered production features end to end** through implementation, testing, code review, and deployment within an enterprise **Agile and Scrum environment**.

ComPro — IBM Platinum Partner

08/2023 – 10/2023

Software Engineering Intern

Istanbul, Turkey

- **Containerized legacy backend environments** using **Docker, CI/CD pipelines, and Linux**, supporting cloud migration proof of concepts, internal development, and client demonstrations.
- **Evaluated IBM Cloud and AWS** for containerized workloads by comparing **compute, storage, orchestration, cost, and performance tradeoffs** to support migration strategy decisions.
- Conducted technical evaluation of **IBM Cloud vs. AWS for containerized workloads**, analyzing **cost/performance trade-offs across compute (EC2 vs. Virtual Servers), storage (S3 vs. Object Storage)**, and orchestration platforms to inform **client migration strategies**.

PROJECTS

Nummoria — AI-Powered Personal Finance SaaS

10/2025 – Present

Full-Stack Engineer & Product Owner

Istanbul, Turkey

nummoria.com • iOS App: TestFlight Release

- **Solely architected, developed, and deployed** a cross platform personal finance SaaS for **Web and iOS** using **Node.js, React, React Native, and MongoDB**, owning the complete product lifecycle from system design to TestFlight release.
- **Engineered modular REST APIs and optimized MongoDB schemas** for transactions, categories, accounts, investment portfolios, and multi currency financial workflows.
- **Integrated OpenAI and Gemini APIs** to analyze user spending patterns and generate contextual personal finance insights and recommendations.
- **Reduced manual transaction entry by approximately 70%** by building an **OCR based ingestion**

pipeline for automated receipt and bank statement parsing.

- **Validated support for 1,000+ concurrent users through load testing**, using connection pooling, database query optimization, and horizontally scalable backend architecture.
- **Implemented secure authentication and session management** using **Google Sign In, Apple Sign In, JWT, and OAuth 2.0**, aligned with App Store security requirements.

Eyehub — TUBITAK Government Research Project 122E085

11/2023 – 06/2024

Backend Engineer

Istanbul, Turkey

- **Built secure Node.js REST APIs** for a **TÜBİTAK funded dyslexia detection application**, supporting mobile synchronization, authentication, real time data submission, and academic research workflows.
- **Designed and deployed AWS infrastructure** using **EC2 and Lambda** to support reliable concurrent research data ingestion.
- **Modeled medical research data in MongoDB** with validation and data integrity controls for academic analysis and research compliance requirements.
- **Collaborated directly with university researchers** to translate clinical and academic requirements into production backend workflows and deliver project milestones.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, C#, Java, SQL

Machine Learning & Data: scikit-learn, Pandas, NumPy, Matplotlib, supervised learning, regression, classification, feature engineering, model evaluation, XGBoost, neural networks, RNNs, Transformers

Generative AI & Agentic Systems: OpenAI and Gemini API integration, prompt engineering, multi-agent workflows, tool-using agents, LLM-as-a-Judge, structured information extraction, scientific reproducibility workflows

Backend & Full-Stack: FastAPI, Node.js, Express.js, React.js, React Native, Angular, ASP.NET Core, REST APIs, OAuth 2.0, JWT, microservices

Databases & Cloud: MongoDB, SQL Server, MySQL, AWS (EC2, Lambda, S3), Docker, CI/CD, Linux, Bash

Engineering & Testing: System design, REST API design, data modeling, caching, Git, Jest, Moq, Postman, Swagger/OpenAPI, unit and integration testing

EDUCATION

B.Sc. Computer Engineering

10/2021 – 08/2025

Bahcesehir University

Istanbul, Turkey

LANGUAGES

- **Turkish** (Native)
- **English** (Full Professional Proficiency)
- **German** (B2 — Certification Prep)